Identifying Alzheimer's Disease through Brain Volumes Longitudinal Data

A study about Alzheimer Neurodegenerative Dementia influence on human brain volumes

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Alzheimer's Disease: The Diagnostic Challenge

Alzheimer's disease is difficult to diagnose early because its symptoms overlap with normal aging and other forms of dementia, making clinical assessments noisy and uncertain.

Pathological brain changes begin years before clear cognitive decline, so by the time symptoms are obvious, substantial and often irreversible damage has already occurred.

Research Question

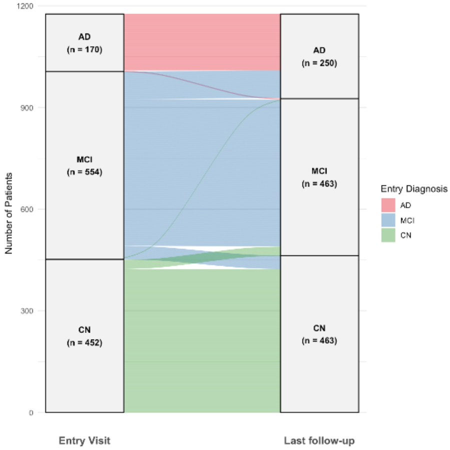
Is it possible to develop a more robust method to identify high-risk patients for Alzheimer's disease using longitudinal brain volumes?

Why?

- Earlier and more accurate interventions
- Better patients stratification for trials and care
- Objective scalable risk assessment

Dataset

- 1176 patients – longitudinal structure
- 117 brain region volumes extracted from MRI images
- 3 diagnostic groups:
  - Cognitive Normal (CN)
  - Mild Cognitive Impairment (MCI)
  - Alzheimer's Disease (AD)
- Additional clinical features (e.g., age, genotype)



Distribution of patients across diagnostic groups

Methodological Roadmap

A. Atrophy Rates (Speed of Neurodegeneration)

1. Brain atrophy estimation
2. Depth based classification
3. Dispersion based classification

B. Functional Analysis (Trajectory Typicality)

1. Functional Envelope-Based Classifier
2. Classification Models Based on MBD
3. Logistic classification

Quantile Regression

$$y_{ij} = \beta_{0,i} + \beta_{1,i} t_{ij} + \varepsilon_{ij}, \qquad i = 1, \dots, I, \;\; j = 1, \dots, n_i$$

$$\begin{aligned} y_{i,j} &: \text{ volume of patient } i \text{ at visit } j, \\ t_{i,j} &: \text{ years from baseline of visit } j \text{ for patient } i, \\ I &: \text{ number of patients,} \\ n_i &: \text{ number of visits for patient } i \end{aligned}$$

$$\text{Atrophy Rate}_i = \frac{\hat{\beta}_{1,i}}{\hat{\beta}_{0,i}} \times 100$$

Projection Depth

MAD Standardization

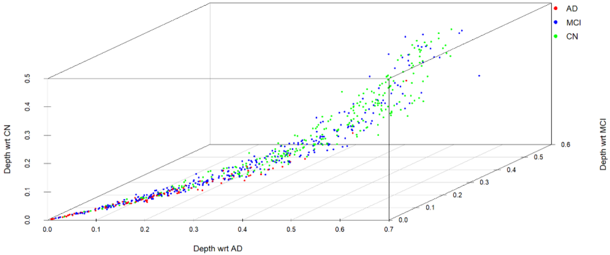
$$Z^{\mathrm{MAD}} = \frac{Z - \mathrm{med}(Z)}{1.4826\ \mathrm{med}(|Z - \mathrm{med}(Z)|)}$$

Projection depth

$$PD(x) = \frac{1}{1 + O(x)}, \qquad O(x) = \sup_{\|u\|=1} \frac{|u^\top x - \mathrm{med}(u^\top X)|}{\mathrm{MAD}(u^\top X)}$$

Depth Space

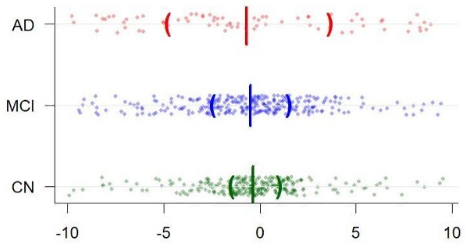
$$f : \mathbb{R}^p \to \mathbb{R}^3, \qquad f(X_i) = \left(D_{\mathrm{CN}}(X_i),\, D_{\mathrm{MCI}}(X_i),\, D_{\mathrm{AD}}(X_i)\right)$$



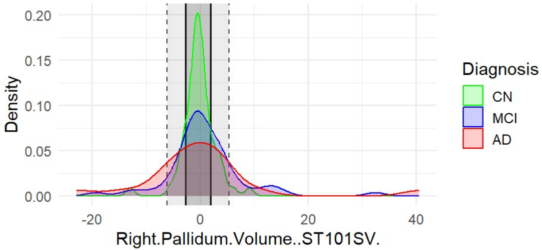
Depth-Based Classification

- DD-α: Accuracy 0.540
- k-NN: Accuracy 0.546
- SVM: Accuracy 0.537
- Inductive Conformal Classification
    - Average prediction set size: 2.69

Closer Inspection of the Data



Distribution of patients across diagnostic groups



Atrophy rate distribution

Dispersion Quantifiers

- Mean Absolute Deviation (CV accuracy: 0.557)
- Variance of Absolute Deviations (0.539)
- Skewness of Absolute Deviations (0.432)
- Correlation-Pattern Deviation (0.600)

Functional Envelope-Based Classifier

- Build healthy envelopes (CN reference)
- Measure patient deviation from CN
- Map to CN / MCI / AD
- Mean Accuracy (5-fold CV): 0.523

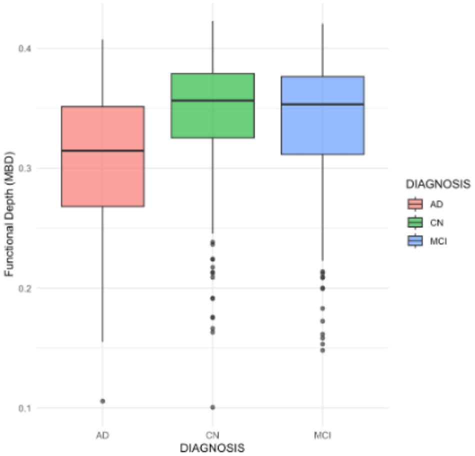
Modified Band Depth (MBD)

$$\text{Depth}_i = \frac{1}{V}\sum_{v=1}^{V} MBD(f_i^{(v)})$$

$$i = 1, \ldots, N,$$
$$v = 1, \ldots, V,$$
$$t = t_1, \ldots, t_L,$$
$$f_i^{(v)}(t) = \text{interpolated trajectory},$$

MDB: modified band depth

Accuracy (5-fold CV): 0.449

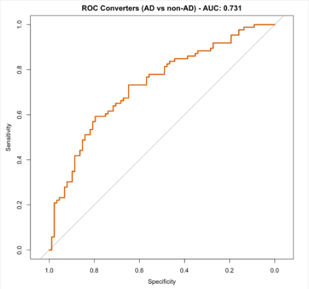


Distribution of patients across diagnostic groups

Logistic Regression Model

$$P(AD) = \text{logit}^{-1}\left(\beta_0 + \beta_1 \cdot depth_i + \beta_2 \cdot age_i + \beta_3 \cdot genotype_i\right)$$

- CV accuracy (stable): 0.8476
- Accuracy (converters): 0.592
- Optimized threshold accuracy: 0.695
- Balanced accuracy: 0.569



Possible Improvements

- Neural networks
- Involve subject expert (neurologist)
- Larger cohort
- More converter patients

Bibliography

Pokotylo et al. (2019) – *Depth and depth-based classification with R package ddalpha*
Lange et al. (2014) – *Fast nonparametric classification based on data depth*
Vencálek (2017) – *Depth-based classification for multivariate data*